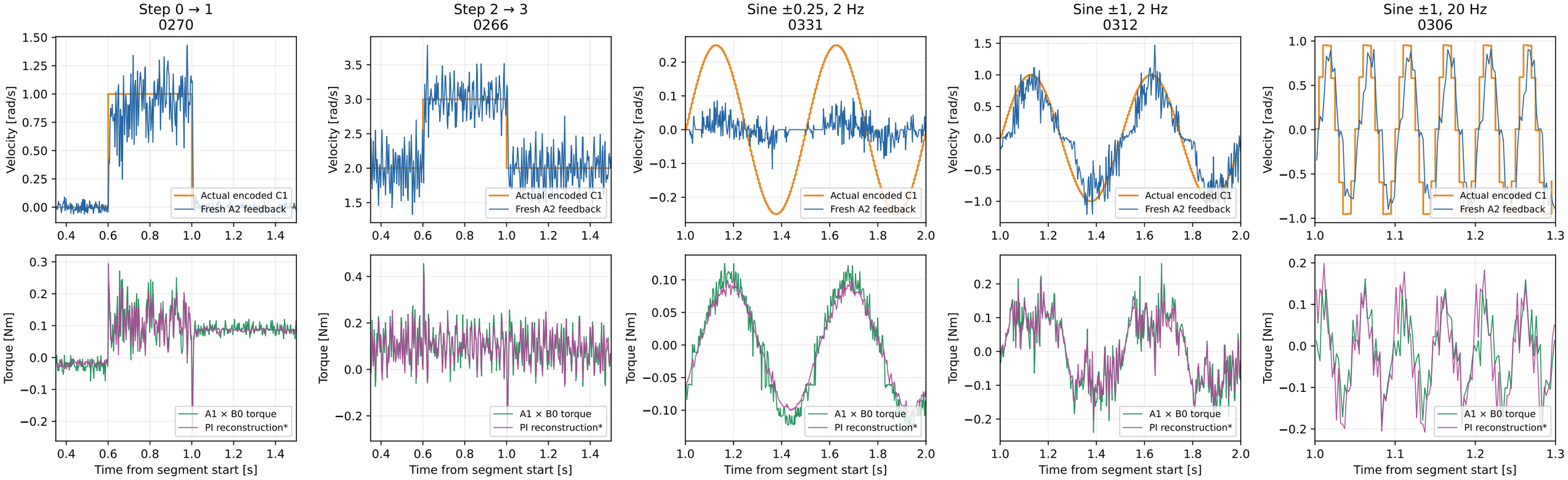

Measured C1 responses — repeat 3; near-zero reversals differ from rotating-wheel steps
Step 0 → 1
0270
Step 2 → 3
0266
Sine ±0.25, 2 Hz
0331
Sine ±1, 2 Hz
0312
Sine ±1, 20 Hz
0306
Velocity [rad/s]
Torque [Nm]
Actual encoded C1
Fresh A2 feedback
A1 × B0 torque
PI reconstruction*
Time from segment start [s]
*Uses observed error/integral, all-data fitted gains and a per-trial offset; this is not a forward simulation. Raw feedback timestamps are unshifted.